

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

MASTER IN COMPUTER APPLICATION (MCA)

THIRD SEMESTER

Scheme of Examination



**DETAILED SYLLABUS
(EFFECTIVE FROM JULY 2018)**

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

MASTER IN COMPUTER APPLICATION (MCA)

THIRD SEMESTER **Scheme of Examination**

S.No	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem Exam	Tests	Assignment/Quiz	End Sem Practical Viva	Practical Record/Assignment/Quiz/Presentation	
1	MCA31	Computer oriented optimization Techniques	3	1	-	4	70	20	10		-	100
2	MCA32	Data Base Management System	3	1	-	4	70	20	10	-	-	100
3	MCA33	Software Engineering	3	1	-	4	70	20	10	-	-	100
4	MCA34	Theory of Computation	3	1	-	4	70	20	10	-	-	100
5	MCA35	Unix and Shell Programming	3	1	-	4	70	20	10	-	-	100
6	MCA36	RDBMS lab	-	-	8	8	-	-	-	100	50	150
7	MCA37	Unix Lab	-	-	2	2	-	-	-	30	20	50
		Total	15	5	10	30	350	100	50	130	70	700

L: Lecture - T: Tutorial - P: Practical

MCA31 - Computer oriented optimization Techniques

UNIT-I

Introduction of operation research. LP Formulations, Graphical method for solving LP's with 2 variables, Simplex method, Duality theory in linear programming and applications, Integer linear programming, dual simplex method.

UNIT-II

Transportation problem, Assignment problem. Dynamic Programming : Basic Concepts, Bellman's optimality principles, Dynamics programming approach in decision making problems, optimal subdivision problem. Sequencing Models: Sequencing problem, Johnson's Algorithm for processing n jobs through 2 machines, Algorithm for processing n jobs through 3 or more machines, Processing 2 jobs through n machines.

UNIT-III

Project Management: PERT and CPM : Project management origin and use of PERT, origin and use of CPM, Applications of PERT and CPM, Project Network, Diagram representation, Critical path calculation by network analysis and critical path method (CPM), Determination of floats, Construction of time chart and resource labelling, Project cost curve and crashing in project management, Project Evaluation and review Technique (PERT).

UNIT-IV

Queuing Models : Essential features of queuing systems, operating characteristics of queuing system, probability distribution in queuing systems, classification of queuing models, solution of queuing M/M/1 : /FCFS, M/M/1 : N/FCFS, M/M/S : /FCFS, M/M/S : N/FCFS

UNIT-V

Inventory Models : Introduction to the inventory problem, Deterministic Models, The classical EOQ (Economic Order Quantity) model, Inventory models with deterministe demands(no shortage & shortage allowed), Inventory models with probabilistic demand, multiitem determinise models.

BOOKS

1. Gillet B.E. : Introduction to Operation Research, Computer Oriented Algorithmic approach - Tata McGraw Hill Publishing Co. Ltd. New Delhi.
2. P.K. Gupta & D.S. Hira, "Operations Research", S.Chand & Co.
3. J.K. Sharma, "Operations Research: Theory and Applications", Mac Millan.
4. S.D. Sharma, "Operations Research", Kedar Nath Ram Nath, Meerut (UP).
5. S.S. Rao "Optimization Theory and Application", Wesley Eastern.
6. Tata Hamdy, A "Operations Research - An Introduction", Fifth Edition, Prentice Hall of India Pvt. Ltd., New Delhi.
7. Taha H.A. "Operations Research an Introduction" McMillan Publication.

MCA 32 - Data Base Management System

UNIT- I

Introduction to database systems Purpose of Database system, View of data, characteristics of Database approach, Architecture for a Database System, Advantages and Disadvantages of DBMS, Database Users and Administrator, Database design and ER model Data Model Classification.

UNIT-II

Structure of relational database, database schema, key, relational operations, formal relational query languages

UNIT-III

Features of good Database design, Universal Relation, Anomalies in a Database, Atomic Domain and 1NF, Functional Dependency Theory, Decomposition Using Functional Dependency, Algorithm For Decomposition, Decomposition Using Multivalued Dependency More Normal Forms, Database Design Process

UNIT-IV

Basic Concepts of Indexing and Hashing , Query Processing , Measures of Query cost , Query processing for select, Sort Join operations. Basics of Query optimization, Transformation of Relational expression, Estimating Statistics of Expression, Choice of evaluation plan

UNIT-V

Transaction Concepts, Features of database transaction, Concurrency control in database Lock base, Time stamp base, validation base Protocols, Database Recovery system

TEXT & REFERENCE BOOKS :

- SILBERSCHATZ KORTH AND SUDARSHAN-DATABASE SYSTEM CONCEPTS, 6TH ED. TATA MC-GRAW HILL.
- RAGHU RAMA KRISHNAN-DATABASE MANAGEMENT SYSTEMS, 2ND ED. TATA MC-GRAW HILL
- RAJESH NARANG – DATABASE MANAGEMENT SYSTEM, 2ND ED. PHI

MCA33 - Software Engineering

UNIT - I

Software : Software Characteristics and Applications, Software Engineering - A Layered Technology, Software Process Models - Linear Sequential Model, Prototype & RAD Model, Incremental Model and Spiral Model. Project Metrics : Software Measurement–Size Oriented, Function Oriented Metrics, Extended Function Point Metrics.

UNIT - II

Software Project Planning: Objectives, Decomposition Techniques, and Empirical Estimation Models. Analysis Concept and Principles: Requirement Analysis, Analysis Principles.

UNIT – III

Design Concepts and Principles: Design Process, Design Concepts, Design Principles, Effective Modular Design, Human Computer Interface Design, Interface Design Guidelines.

UNIT - IV

S/W Quality Assurance : Quality Concepts, Reliability S/W Testing Models : S/W Testing Fundamentals, White and Black Box Testing, Basic Path Testing, Testing Strategies : Strategic Approach to S/W Testing, Unit Testing, Integration Testing, Validation Testing, System Testing

UNIT - V

S/W Reuse : Reuse Process, Classification and Retrieving Components, Economics of S/W Reuse CASE : Introducing to CASE, Taxonomy of Case Tools

TEXT & REFERENCE BOOKS :

- SOFTWARE ENGINEERING BY R.S.PRESSMAN
- AN INTEGRATED APPROACH TO SOFTWARE ENGINEERING BY PANKAJ JALOTE

MCA 34 - Theory of Computation

UNIT-I

Review of Mathematical Preliminaries : Set, Relations and functions, Graphs and trees, string, alphabets and languages. Principle of induction, predicates and propositional calculus. Theory of Automation : Definition, description, DFA,NFA, Transition systems,2DFA, equivalence of DFA & NDFA, Regular expressions, regular grammar, FSM with output (mealy and moore models), Minimisation of finite automata.

UNIT-II

Formal Languages : Definition & description, Parse structured grammars & their classification, closure properties of families of language, regular grammar, regular set & their closure properties, finite automata, equivalence of FA and regular expression, equivalence of two way finite automata, equivalence of regular expressions.

UNIT -III

Context-Free grammar & PDA : Properties unrestricted grammar & their equivalence, derivation tree simplifying CFG, unambiguifying CFG, λ -productions, normal form for CFG, Pushdown automata, 2way PDA, relation of PDA with CFG, Determinism & Non determinism in PDA & related theorems, parsing and pushdown automata.

UNIT-IV

Turing Machine : Model, design, representation of TM, language accepted by TM, universal turing machine, determine & non-determinism in TM, TM as acceptor/generator/algorithms, multidimensional, multitape, multitape, Two way infinite tape, multihead, Halting problems of TM.

UNIT-V

Computability : Concepts, Introduction to complexity theory, Introduction to undecidability, recursively enumerable sets, primitive recursive functions, recursive set, partial recursive sets, concepts of linear bounded Automata, context sensitive grammars & their equivalence.

BOOKS

1. Hopcroft & Ullman "Introduction to Automata theory, languages & Computation" , Narosa Publishing house.
2. Lewis Papadimitra "Theory of Computation" , Prentice Hall of India, New Delhi.
3. Peter linz, "An Introduction to formal language and automata", Third edition, Narosa publication

MCA35 - Unix and Shell Programming

Unit-I

General Overview of the System: System structure, user perspective, O/S services assumption about Hardware The Kernel and buffer cache architecture of Unix O/S, System concepts, Kernel data Structure, System administration, Buffer headers, Structure of the buffer pool, Scenarios for retrieval of the buffer, Reading and writing disk block, Advantage and disadvantage of buffer cache.

Unit-II

Internal Representation of Files: INODES, Structure of regular, Directories conversions of a path name to an inode, Super block, Inode assignment to a new file, Allocation of disk blocks.

System Calls for the System: Open read write file and record close, File creation, Operation of special files change directory and change root, change owner and change mode, STAT and FSTAT, PIPES Mounting and unmounting files system, Link Unlink.

Unit-III

Structures of Processes and process control: Process states and transitions layout of system memory, the context of a process, manipulation of process address space, Sleep process creation/termination. The user Id of a process, changing the size of a process. The SHELL

Interprocess Communication and multiprocessor system: Process tracing system V IPO network communication sockets problem of multiprocessors systems, solution with master and hare process, and solution with semaphores.

Unit-IV

Introduction to shell scripts: shell Bourne shell, C shell, Unix commands, permissions, editors, filters sed, grep family, shell variables, scripts, metacharacters and environment, if and case statements, for while and until loops. Shell programming.

Unit-V

Awk and perl Programming: Awk pattern scanning and processing language, BEGIN and END patterns, Awk arithmetic and variables, Awk built in variable names and operators, arrays, strings, functions, perl; the chop() function, variable and operators, \$_ and \$. , Lists, arrays, regular expression and substitution, file handling, subroutines, formatted printing.

Linux:

History & Features of Linux, Linux structure, various flavours of linux.

Books

1. M.J. Bach “Design of UNIX O.S. “, Prentice Hall of India